SERANGOON JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 1

CANDIDATE NAME [ ]

CLASS [ ]

CHEMISTRY — 8873/02

JC2 Preliminary Examination — **12 September 2018**
Paper 2 Structured Questions — **2 hours**

Candidates answer on the Question Paper.
Additional Materials: Data Booklet

READ THESE INSTRUCTIONS FIRST

Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** the questions.

Section B
Answer **one** question.

The use of an approved scientific calculator is expected, where appropriate.
A Data Booklet is provided.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [ ] at the end of each question or part questions.

For Examiner's Use	
1	/17
2	/9
3	/8
4	/12
5	/14
6 / 7	/20
TOTAL	/ **80**

This document consists of **23** printed pages and **1** blank page.

Section A

Answer all the questions in this section in the spaces provided.

1 Combustion data can be used to calculate the empirical formula, molecular formula and relative molecular mass of many organic compounds.

(a) Define the term *relative molecular mass.*

....

....

....
[1]

(b) **B** is an alcohol, C_xH_yO.

A 20 cm^3 gaseous sample of **B** was completely burnt in 200 cm^3 of oxygen (an excess). The final volume, measured under the same conditions as the gaseous sample, was 250 cm^3.

Under these conditions, all water present was vaporised. Removal of the water vapour from the gaseous mixture decreased the volume to 170 cm^3. The remaining gaseous mixture was treated with concentrated alkali, and the eventual volume was decreased to 110 cm^3.

Given the equation for the complete combustion of **B**, answer the questions below.

$$C_xH_yO + zO_2 \rightarrow xCO_2 + \frac{y}{2}H_2O$$

Calculate the value of ***x*** and ***y***.

[2]

(c) Period 3 elements also react with oxygen to form oxides.

Sodium oxide and sulfur dioxide are two such oxides.

(i) Write chemical equations, with state symbols, when each of the above oxides react with water.

sodium oxide …………………………………………………………………………

sulfur dioxide …………………………………………………………………………

[2]

(ii) State and explain the pattern of change of oxidation number which occurs to both oxygen and the Period 3 elements (sodium to silicon) when they react together.

....

....

....

[2]

(d) The first ionisation energies of elements across Period 3 show a general increase.

Aluminium and sulfur do **not** follow this general trend.

(i) Write an equation to define the first ionisation energy of aluminium.

....

[1]

(ii) Explain the general increase in ionisation energy across Period 3.

....

....

....

[2]

(iii) Explain why aluminium has a lower first ionisation energy than magnesium.

....

....

[1]

(e) Silicon is an element in Period 3 displaying unique properties such as electrical conductivity at high temperatures and high melting point.

(i) Complete the sketch of the melting point trend for Period 3 elements in **Figure 1.1**.

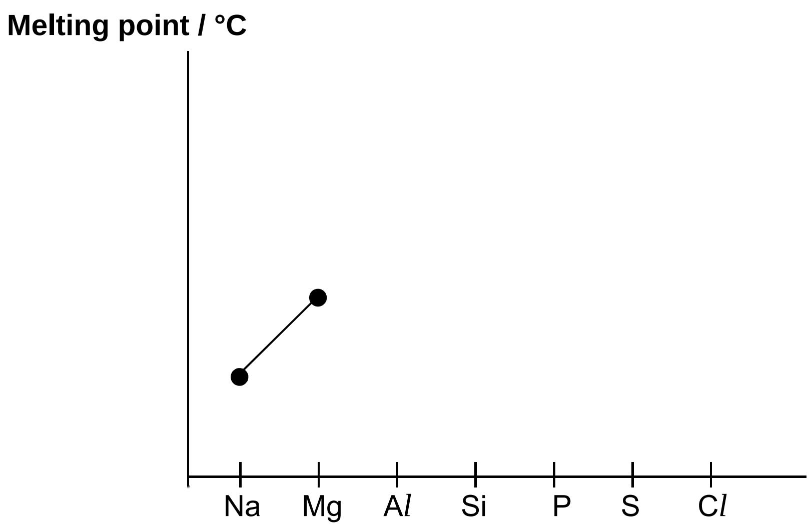

Figure 1.1

[1]

(ii) With the aid of a well-labelled diagram, explain the abnormally high melting point of silicon, using concepts of structure and bonding.

....

....

....

....

[3]

(iii) When silicon reacts with magnesium, Mg_2Si forms. Mg_2Si is thought to contain the Si^{4-} ion.

State the full electronic configuration of Si^{4-}.

$1s^2$…………………………………………… [1]

(iv) Solid Mg_2Si reacts with dilute hydrochloric acid to form gaseous SiH_4 and a solution of magnesium chloride.

Write an equation to show the reaction described above. Include state symbols.

.... [1]

[Total: 17]

2 Hydrazine has many industrial uses such as rocket fuel, pesticides and to prepare gas precursors used in air bags. Liquid hydrazine undergoes combustion according to the following equation:

$$N_2H_4\,(l) + O_2\,(g) \rightarrow N_2\,(g) + 2H_2O\,(l)$$

A chemist conducted an experiment to determine the standard enthalpy change of combustion of hydrazine where 0.420 g of hydrazine was burnt to heat up a beaker containing 200 cm^3 of water. The temperature of water in the beaker was recorded as follows:

Initial temperature /°C	29.3
Final temperature / °C	37.4

(a) **(i)** Draw the dot-and-cross diagram of hydrazine.

[1]

(ii) Explain what is meant by *standard enthalpy change of combustion of hydrazine.*

....

....
[1]

(iii) Calculate the standard enthalpy change of combustion of hydrazine.

You may assume the process has 80 % efficiency.

[2]

(iv) Given the following data:

enthalpy change of formation of steam = −242 kJ mol^{-1}
enthalpy change of vapourisation of water = +44 kJ mol^{-1}

Using the value you have calculated in **(a)(iii),** complete the following energy cycle to determine the standard enthalpy of formation of hydrazine.

$$\begin{tikzcd} N_2\,(g) + 2H_2\,(g) \arrow[r] \arrow[d] & N_2H_4\,(l) \arrow[d] \\ N_2\,(g) + 2H_2O\,(g) & N_2\,(g) + 2H_2O\,(l) \arrow[l] \end{tikzcd}$$

[2]

(b) **(i)** The standard enthalpy change of formation of hydrazine gas is +235 kJ mol^{-1}.

Using appropriate data from the *Data Booklet*, calculate the bond energy of N−H in hydrazine gas.

[2]

(ii) Suggest a reason for the difference in the N−H bond energy value obtained from **(b)(i)** with the value given in the *Data Booklet*.

....

....
[1]

[Total: 9]

3 This question is about nanoparticles and its applications.

(a) Finely divided nanoparticles of nickel are found to be extremely good heterogeneous catalyst for many organic synthesis reactions.

(i) Define the term *catalyst*.

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....
[1]

(ii) State the difference between a *nanoparticle* and a *nanomaterial*.

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....
[1]

(iii) Briefly explain how nickel nanoparticles operate as heterogeneous catalyst.

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....
[2]

(b) One study found that 85 % of sunscreens available in Singapore are likely to contain titanium dioxide nanoparticles due to its ability to block UV radiation while remaining transparent on the skin.

Although the health risk posed by these sunscreen creams is minimal, this is not so for aerosol spray sunscreens. Suggest a reason why aerosol spray sunscreens pose a greater health risk.

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....

[1]

(c) Geckoes and many insects have adopted nanoscale fibrillary structures on their feet as adhesion devices. Many have the remarkable ability to run at any orientation on just about any smooth or rough, wet or dry, clean or dirty surface.

Using your understanding of molecular forces, suggest why a gecko can support its own weight and stick to the wall.

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....

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[3]

[Total: 8]

4 Acids are an important component in both winemaking and the finished product of wine. They have a direct influence on the colour and taste of the wine. The measure of the amount of acid in wine is known as the "total acidity".

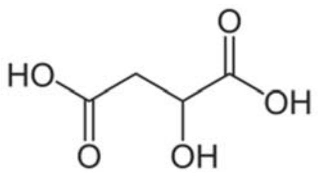
A label on a bottle of wine describes it as having a "total acidity" of 5.8 g dm^{-3}.

(a) **(i)** The acids found in wine behave as *Brønsted-Lowry acids*. Define a *Brønsted-Lowry acid*.

....

[1]

(ii) One of the most common acid present in wine is malic acid.



malic acid
M_r = 134.0
K_a = 3.98 x 10^{-4} mol dm^{-3}

Assuming that malic acid behaves effectively as a monobasic acid, RCOOH. Write an equation to show the dissociation of malic acid.

....

[1]

(iii) Calculate the concentration in, mol dm^{-3}, of malic acid and the pH of the wine.

[2]

(iv) The dissociation of malic acid is endothermic. Explain the impact on its pH **and** K_a when temperature increases.

....

....

....

.... [3]

(v) During the ingestion of wine, mouth saliva interacts with it through a buffering action involving CO_3^{2-} / HCO_3^-. Use an equation to illustrate how the buffer system in saliva maintains pH when wine is consumed.

.... [1]

(b) Malic acid undergoes elimination with concentrated sulfuric acid.

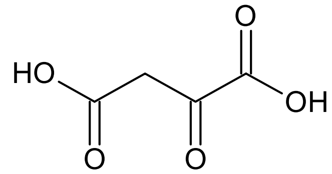
(i) Write a balanced equation for the reaction that took place.

.... [1]

(ii) Draw two possible isomers formed in **(b)(i)** and state the type of isomerism.

[2]

(c) Suggest the reagent required for the synthesis of malic acid from the following compound.

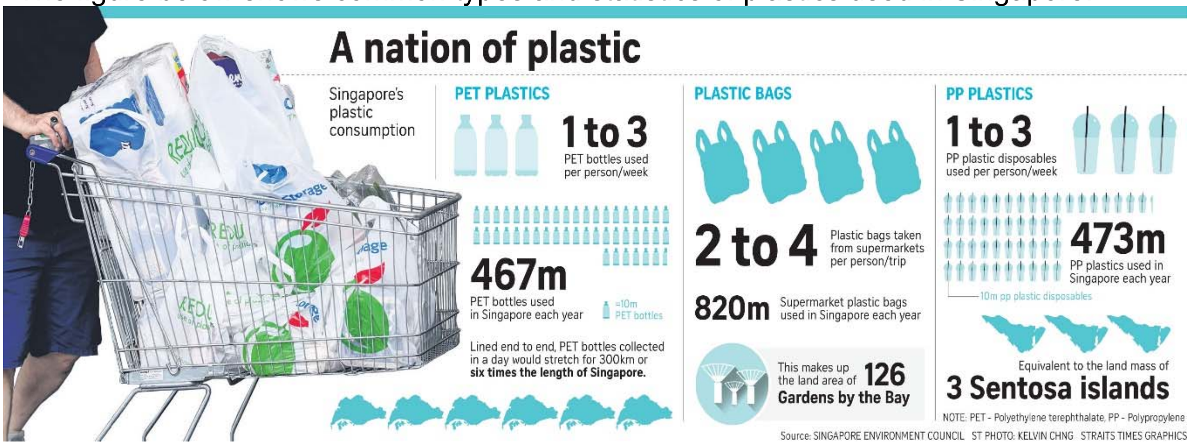


.... [1]

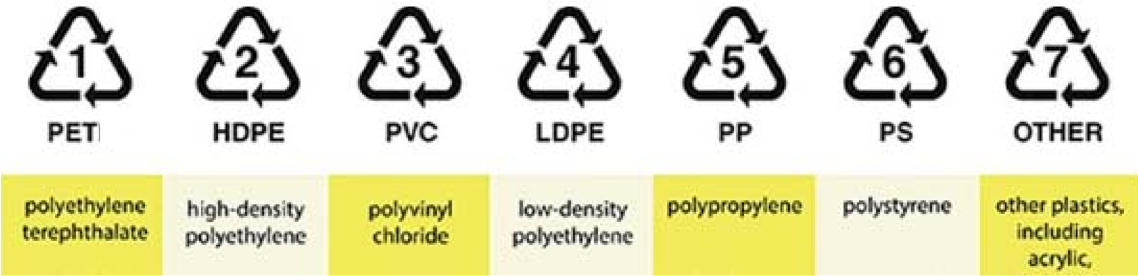
[Total: 12]

5 Singapore uses almost one plastic item per person per day, but fewer than 20% of these are recycled, according to the Singapore Environment Council (SEC). According to the National Environment Agency (NEA), only 6% by mass of plastics are recycled.

The figure below shows common types and statistics of plastics used in Singapore:



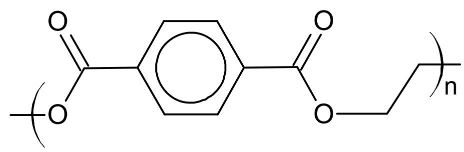
The ASTM International Resin Identification Coding System, often abbreviated as the RIC, is a set of symbols appearing on plastic products that identify the plastic resin out of which the product is made. Understanding the seven plastic codes will make it easier to choose plastics and to know which plastics to recycle:



The methods to reduce on waste in Singapore follows the three "Rs" – Reduce, Reuse and Recycle. However, not all plastics can be reused or recycled.

According to some environmental watch groups, in order for a material to be reused, especially with applications on food and drink, it should not leach compounds that are toxic to human consumption (e.g. benzene derivatives). In order for a material to be recycled, it must be able to be melted and remoulded under high heat, and not decompose or char.

(a) **(i)** PET polymers are thermoplastics and are observed to have the repeat unit shown below.

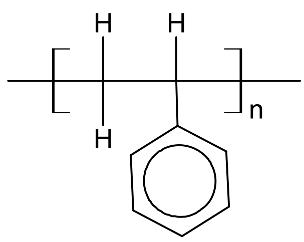


Draw the monomers of PET and suggest the classification of this polymerisation.

Class of polymerisation: ……………………………………..

[2]

(ii) PS polymers are thermosets and are observed to have the repeat unit shown below.



Draw the monomer of PS and suggest the classification of this polymerisation.

Class of polymerisation: ……………………………………..

[2]

(iii) With the aid of a simple well-labelled diagram, explain how thermoplastics differ from thermosets in their bonding between polymer chains.

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....

.... [4]

(iv) State if PET and PS can be recycled or reused.

....

.... [1]

(b) Explain the relative rigidity and state possible uses of both LDPE and HDPE.

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.... [3]

(c) A school performed a survey on 1000 students and 200 staff members. The survey found that the average person in school would purchase a drink in a plastic cup, and use two pieces of plastic crockery, for lunch every day. Each of those items used had a symbol emblazoned at the base shown below.

Identify the polymer used, and calculate the percentage waste contributed by this school in Singapore in a year in relation to the type of polymer resin used.

Polymer used: …………………………….

[2]

[Total: 14]

Section B

Answer **one** question from this section in the spaces provided.

6 (a) Transition metals are used in many commercial applications such as catalysts. The manufacture of sulfuric acid via the Contact process uses vanadium(V) oxide as catalyst.

Stage one of this process involves the conversion of sulfur dioxide into sulfur trioxide.

$$2SO_2\,(g) + O_2\,(g) \rightleftharpoons 2SO_3\,(g) \qquad \Delta H = -196 \text{ kJ mol}^{-1}$$

2.00 moles of $SO_2(g)$ and 2.00 moles of $O_2(g)$ are sealed in a 40 dm^3 container with the addition of catalyst, at constant temperature and pressure. The resulting equilibrium contains 1.98 moles of $SO_3(g)$.

(i) Use the information above to calculate the K_c value for the reaction, stating its units clearly.

[3]

(ii) State and explain the effect of increasing pressure on the **rate** of production of SO_3.

....

....

....

[2]

(iii) State and explain the effect of increasing pressure on the **yield** of SO_3.

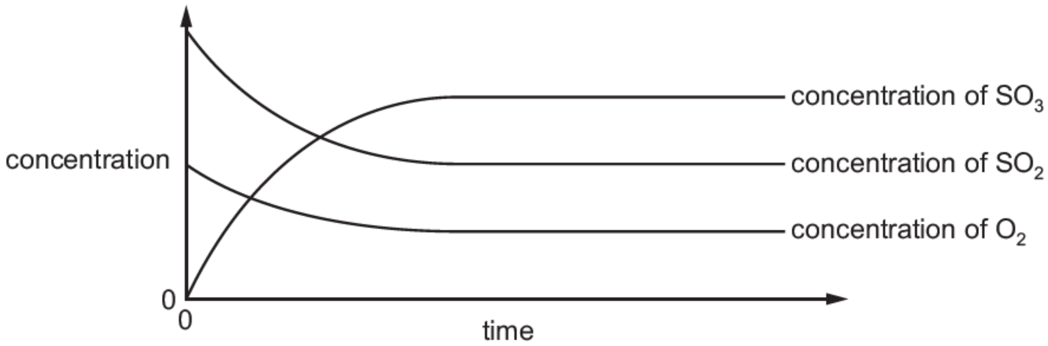
....

....

.... [2]

(iv) Another experiment of stage one was carried out.

The following graph showing how the concentrations of the three species in the system change with time was sketched.



Study the graph and explain the following:

(I) the gradients of the SO_2 and O_2 lines decrease with time

....

....

....

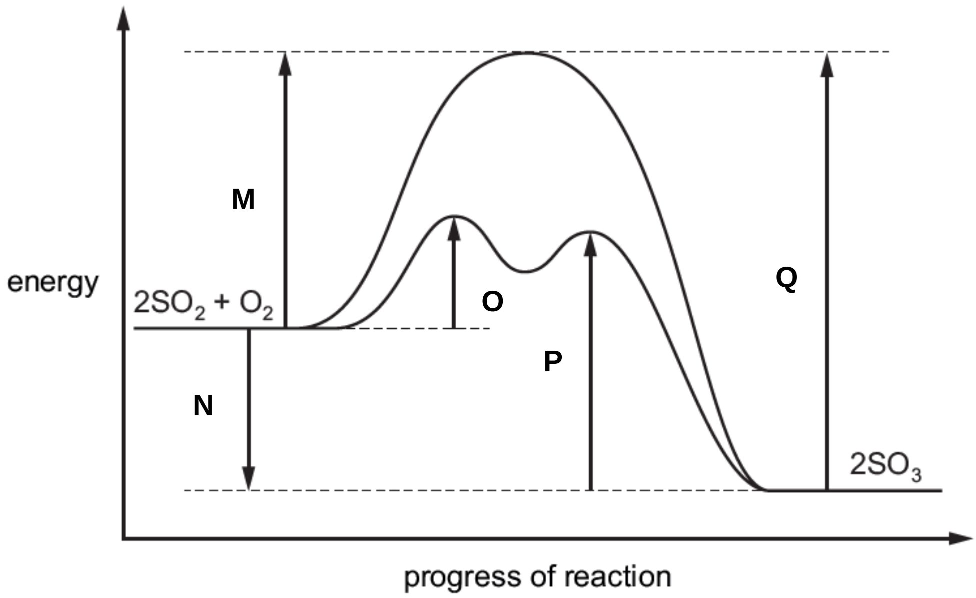
(II) the initial gradient of the SO_2 line is steeper than that of the O_2 line

....

....

.... [2]

(v) A reaction pathway diagram for both the catalysed and uncatalysed reactions of stage one is shown.



The letters **M – Q** represent energy changes.

Complete the table by stating which letter, **M – Q**, represents the energy change described.

energy change	letter
the energy change for the production of SO_3	
the activation energy for the production of SO_3 in the absence of a catalyst	
the activation energy for the first step in the **decomposition** of SO_3 in the presence of a catalyst	

[3]

(b) Transition metal manganese present in strong oxidising agents such as potassium manganate(VII), are used in many organic chemistry applications.

Butan-2-ol was heated with acidified potassium manganate(VII) to produce **R**.

$$CH_3CH(OH)CH_2CH_3 \xrightarrow{\textbf{I}} \underset{\textbf{R}}{CH_3COCH_2CH_3}$$

(i) Write a balanced equation for reaction **I**.

....

[1]

(ii) There are two constitutional isomers of **R**, which are also carbonyl compounds. In the boxes below, draw the skeletal formulae of the isomers of **R**.

[2]

(iii) Suggest an alternative reagent and condition for reaction **I**.

....

[1]

(c) Mohr's salt, $(NH_4)_2Fe(SO_4)_2{\cdot}xH_2O$, is a hydrated form of ammonium iron(II) sulfate, where ***x*** represents the number of moles of water in 1 mole of the salt.

A student wanted to determine the value of ***x***.

0.784 g of the hydrated salt was dissolved in water and 10 cm^3 of sulfuric acid was added. All of the solution was titrated with 0.0200 mol dm^{-3} potassium manganate(VII) where 20.00 cm^3 of potassium manganate(VII) solution was required for the complete reaction with the Fe^{2+} ions.

(i) Use changes in oxidation number, or otherwise, balance the equation for the reaction taking place.

MnO_4^- (aq) + ….Fe^{2+} (aq) + ….H^+ (aq) → ….Mn^{2+} (aq) + ….Fe^{3+} (aq) + ….H_2O (*l*)

[1]

(ii) Calculate the amount of Fe^{2+} in the sample of the salt.

[1]

(iii) Calculate the relative formula mass of $(NH_4)_2Fe(SO_4)_2{\cdot}xH_2O$ and hence, determine the value of ***x***.

[2]

[Total: 20]

7 (a) Samples of 2-bromopropane were dissolved in dilute aqueous ethanol and reacted with potassium hydroxide solution. Several experiments were carried out at constant temperature. The initial rate of reaction was determined in each case.

Expt	$[(CH_3)_2CHBr]$ / mol dm^{-3}	$[OH^-]$ / mol dm^{-3}	Rate / mol dm^{-3} s^{-1}
1	0.015	0.015	13.5
2	0.015	0.030	13.5
3	0.030	0.045	27.0

(i) Determine the order of reaction with respect to 2-bromopropane and potassium hydroxide.

[2]

(ii) Hence, write the rate equation and determine the rate constant, indicating the units clearly.

[2]

(iii) In another experiment, 2-bromopropane was reacted **hot aqueous** potassium hydroxide solution instead.

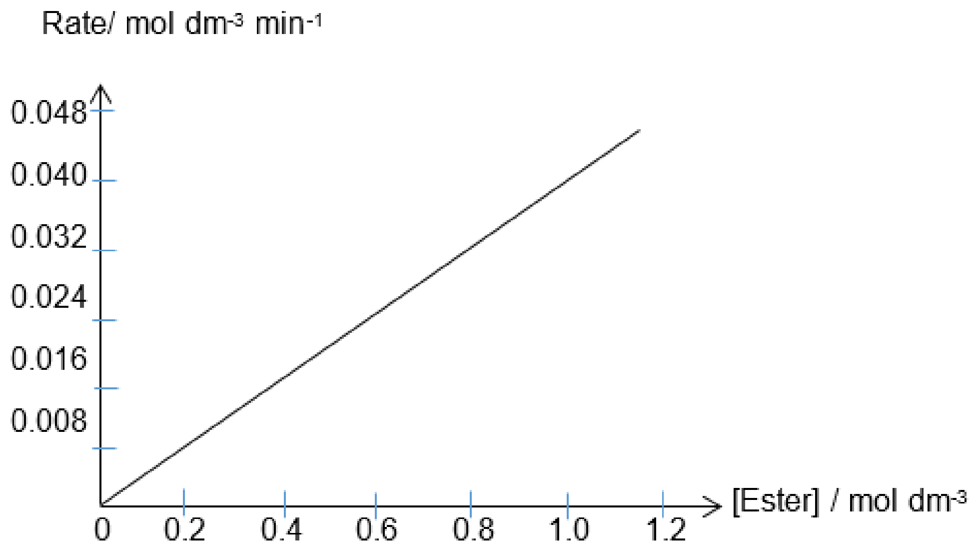
Write a balanced equation for the reaction that occurs.

....

[1]

(b) A series of experiments was conducted to investigate the kinetics of the hydrolysis of an ester.

The graph below shows the results obtained when concentrations of the ester were varied with the addition of large quantity of water at 400 K. Order of reaction with respect to the ester was found to be one.



(i) Using the graph, determine the value of rate constant, *k*.

[1]

(ii) Predict how the gradient of the graph would change at 500 K. Explain your answer.

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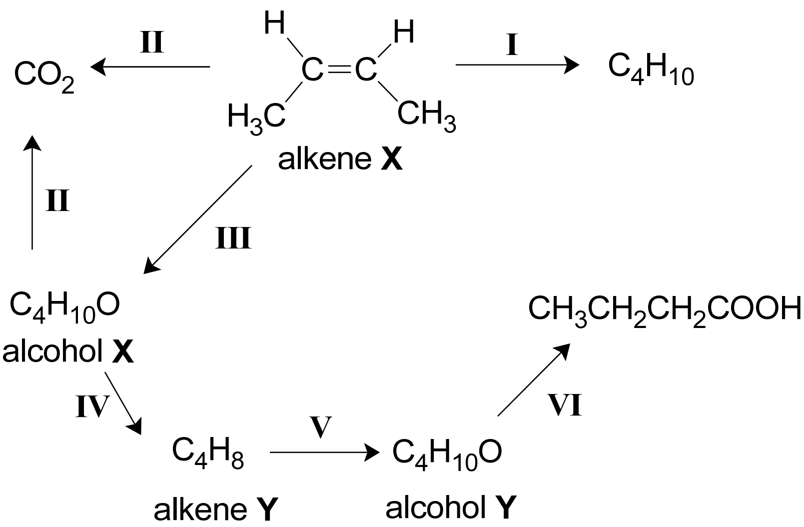
....
[3]

(c)



(i) Construct a balanced chemical equation for reaction **II** involving alkene **X.**

.... [1]

(ii) When alkene **X** undergoes reaction with water in the presence of an acid catalyst, alcohol **X**, a secondary alcohol, is produced.

Draw the displayed formula of alcohol **X**. Hence, name the carbonyl compound that can be reduced to form this alcohol.

[2]

(iii) Draw the structural formula of alkene **Y** and alcohol **Y**.

[2]

(iv) Acidified potassium manganate(VII) was used in the oxidation process of alcohol **Y** to butanoic acid, as seen in reaction **VI**.

The oxidation half-equation for the reaction is shown below.

$$C_4H_{10}O + H_2O \rightarrow CH_3CH_2CH_2COOH + 4H^+ + 4e$$

With an aid of a *Data Booklet*, construct an overall balanced chemical equation for reaction **VI**, and determine the volume of 1.25 mol dm^{-3} potassium manganate(VII) required to oxidise 15 g of alcohol **Y**.

[3]

(v) Platinum acts as catalyst in reaction **I**. With an aid of a Maxwell-Boltzmann distribution diagram, explain how platinum increases the rate of reaction.

....

....

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[3]

[Total: 20]

END

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